

C958

Calculus I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An analyst at a bakery that opened in 2015 builds a spreadsheet model with 500 rows that uses $f(x) = (x^2 - 9) / (x - 3)$. What is the limit of $f(x)$ as x approaches 3?

- A. 3
- B. 6
- C. It does not exist, because $f(3)$ is undefined.
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A drone maker in Oregon models battery drain with $f(x) = 4x^3 - 5x + 7$. The model was reviewed by two engineers. What is $f'(2)$?

- A. 43
- B. 91
- C. 50
- D. 29

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An engineer at a battery startup with 40 employees models output with $y = (3x + 1)^5$. What is dy/dx ?

- A. $5(3x + 1)^4$
- B. $3(3x + 1)^4$
- C. $(3x + 1)^6 / 18$
- D. $15(3x + 1)^4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A test cart's position in meters is $s(t) = t^3 - 6t^2 + 9t$, measured by a sensor sampling 100 times per second. What is its velocity at $t = 2$ seconds?

- A. -3 m/s
- B. 1 m/s
- C. 2 m/s
- D. 0 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

For $s(t) = t^3 - 6t^2 + 9t$, at what times is the cart at rest?

- A. $t = 1$ only
- B. $t = 0$ and $t = 3$
- C. $t = 1$ and $t = 3$
- D. $t = 2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A farm that has grown organic kale since 2012 has 400 m of fencing to enclose a rectangle along a straight river; no fence is needed on the river side. Fencing costs \$8 per meter. What dimensions give the largest area?

- A. 133.3 m on each of the three sides, area about 17,778 m^2
- B. 100 m by 200 m (200 m along the river), area 20,000 m^2
- C. Whatever keeps cost under \$3,200
- D. 100 m by 100 m, area 10,000 m^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

At $x = c$, $f'(c) = 0$ and $f''(c) = -4$. What does f have at c ?

- A. A local minimum
- B. The absolute maximum on every interval
- C. An inflection point
- D. A local maximum

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Water flows into a tank at a treatment plant serving 50,000 residents at a rate of $r(x) = 3x^2 + 1$ thousand liters per hour. How much water enters from $x = 0$ to $x = 2$ hours?

- A. 13 thousand liters
- B. 8 thousand liters
- C. 12 thousand liters
- D. 10 thousand liters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

$\int_0^k 2x \, dx = 25$, with $k > 0$. What is k ?

- A. 25
- B. 12.5
- C. 5
- D. About 3.54

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A fitness app built by a team of six has 200 users and grows by $dy/dt = 0.05y$ (t in days). A marketing push is planned for next quarter. About how many users will it have after 10 days?

- A. About 330
- B. 210
- C. About 326
- D. 300

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A quantity following $dy/dt = ky$ doubles every 10 years. What is k ?

- A. 0.2 per year
- B. About 0.0693 per year
- C. 0.1 per year
- D. About 0.693 per year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A signal model at a hearing-aid company, tested on 30 volunteers, uses $f(x) = x^2 \sin x$. What is $f'(x)$?

- A. $2x \sin x$
- B. $2x \cos x$
- C. $2x \sin x + x^2 \cos x$
- D. $2x \sin x - x^2 \cos x$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

An analyst ~~at a bakery that opened in 2015~~ builds a spreadsheet model ~~with 500 rows~~ that uses $f(x) = (x^2 - 9) / (x - 3)$. What is the limit of $f(x)$ as x approaches 3?

Target word: limit at $x = 3$ **Decides:** Factor: $(x - 3)(x + 3)/(x - 3) = x + 3$.

A	Wrong stem number	The point x approaches, not the limit.
B	Correct	After canceling, $x + 3 \rightarrow 6$.
C	Confusion set	The limit can exist even where f is undefined.
D	Dropped element	Reads $0/0$ as 0 without simplifying.

2. Answer A Concept 2

A drone maker ~~in Oregon~~ models battery drain with $f(x) = 4x^3 - 5x + 7$. ~~The model was reviewed by two engineers.~~ What is $f'(2)$?

Target word: derivative at $x = 2$ **Decides:** $f'(x) = 12x^2 - 5$.

A	Correct	$12(4) - 5 = 43$.
B	Inverted operation	Multiplies by 3 but does not lower the exponent: $12(8) - 5$.
C	Dropped element	Keeps the constant 7, whose derivative is 0.
D	Confusion set	That is $f(2)$, not $f'(2)$.

3. Answer D Concept 3

An engineer ~~at a battery startup with 40 employees~~ models output with $y = (3x + 1)^5$. What is dy/dx ?

Target word: derivative of a composite **Decides:** Outer $5u^4$ times inner derivative 3.

A	Dropped element	Forgets the inner derivative.
B	Confusion set	Uses only the inner derivative as the coefficient.
C	Inverted operation	That is the antiderivative.
D	Correct	$5(3x + 1)^4 \times 3$.

4. Answer A Concept 4

A test cart's position in meters is $s(t) = t^3 - 6t^2 + 9t$, ~~measured by a sensor sampling 100 times per second.~~ What is its velocity at $t = 2$ seconds?

Target word: $v(2)$ **Decides:** $v(t) = 3t^2 - 12t + 9$.

A	Correct	$12 - 24 + 9 = -3$.
B	Adjacent category	Average velocity over 0 to 2, not instantaneous.
C	Wrong stem number	That is $s(2)$, the position.
D	Confusion set	That is the acceleration, $6t - 12$, at $t = 2$.

5. Answer C Concept 4 • transfer

For $s(t) = t^3 - 6t^2 + 9t$, at what times is the cart at rest?

Target word: solve $v(t) = 0$ **Decides:** $3t^2 - 12t + 9 = 3(t - 1)(t - 3)$.

A	Dropped element	Misses the second root.
B	Adjacent category	Where position is zero, not velocity.
C	Correct	Both roots of $v(t) = 0$.
D	Confusion set	That is where acceleration is zero.

6. Answer B Concept 5

A farm ~~that has grown organic kale since 2012~~ has 400 m of fencing to enclose a rectangle along a straight river; no fence is needed on the river side. ~~Fencing costs \$8 per meter.~~ What dimensions give the largest area?

Target word: optimal dimensions **Decides:** $x + 2y = 400$; $A = y(400 - 2y)$; $A' = 0$ at $y = 100$.

A	Confusion set	Equal sides do not maximize area here.
B	Correct	$400 - 4y = 0$ gives $y = 100$, $x = 200$.
C	Camouflage	Cost is fixed by the 400 m; it does not decide the shape.
D	Dropped element	Fences all four sides; drops the river condition.

7. Answer D Concept 5 • transfer

At $x = c$, $f'(c) = 0$ and $f''(c) = -4$. What does f have at c ?

Target word: classify the critical point **Decides:** $f'' < 0$ means concave down.

A	Inverted operation	That needs $f'' > 0$.
B	Absolute language	The test is local only.
C	Confusion set	Needs f'' to change sign, not be negative.
D	Correct	Concave down at a critical point is a peak.

8. Answer D Concept 6

Water flows into a tank ~~at a treatment plant serving 50,000 residents~~ at a rate of $r(x) = 3x^2 + 1$ thousand liters per hour. How much water enters from $x = 0$ to $x = 2$ hours?

Target word: definite integral **Decides:** Antiderivative $x^3 + x$ from 0 to 2.

A	Confusion set	That is $r(2)$, the rate at the end.
B	Dropped element	Leaves out the $+x$ term.
C	Inverted operation	Differentiates: $r'(2) = 6 \times 2$.
D	Correct	$(8 + 2) - 0 = 10$.

9. Answer C Concept 6 • transfer

$\int_0^k 2x \, dx = 25$, with $k > 0$. What is k ?

Target word: solve for upper bound **Decides:** $k^2 = 25$.

A	Wrong stem number	The value of the integral.
B	Inverted operation	Divides 25 by 2 instead of integrating.
C	Correct	The antiderivative is x^2 , so $k^2 = 25$.
D	Confusion set	Uses $2x^2$ as the antiderivative, forgetting to divide by the new exponent: $2k^2 = 25$.

10. Answer A Concept 7

A fitness app ~~built by a team of six~~ has 200 users and grows by $dy/dt = 0.05y$ (t in days). ~~A marketing push is planned for next quarter.~~ About how many users will it have after 10 days?

Target word: $y(10)$ **Decides:** $y = 200e^{(0.5)}$.

A	Correct	$200 \times e^{0.5} \approx 200 \times 1.649 \approx 330$.
B	Dropped element	Applies the rate for only one day.
C	Confusion set	Daily compounding 200×1.05^{10} , not continuous growth.
D	Adjacent category	Linear growth: $200 \times (1 + 0.05 \times 10)$.

11. Answer B Concept 7 · transfer

A quantity following $dy/dt = ky$ doubles every 10 years. What is k ?

Target word: solve for k **Decides:** $e^{(10k)} = 2$.

A	Inverted operation	$2 \div 10$ divides the growth factor instead of taking its log.
B	Correct	$k = \ln 2 \div 10$.
C	Adjacent category	$1 \div 10$ treats it as linear.
D	Dropped element	Forgets to divide $\ln 2$ by 10.

12. Answer C Concept 8

A signal model at a hearing-aid company, ~~tested on 30 volunteers,~~ uses $f(x) = x^2 \sin x$. What is $f'(x)$?

Target word: derivative of a product **Decides:** Product rule.

A	Dropped element	Misses the fg' term.
B	Confusion set	Multiplies the two derivatives.
C	Correct	$f'g + fg'$.
D	Inverted operation	Borrows the minus sign from the quotient rule.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Limits of 0/0 forms	A limit describes what $f(x)$ approaches, not f at the point; factor and cancel to resolve 0/0.	Saying the limit does not exist because $f(a)$ is undefined.	Plugging in gives 0/0.
2	Power rule	$d/dx [x^n] = n \cdot x^{n-1}$, and the derivative of a constant is 0.	Keeping the constant or not lowering the exponent.	A polynomial and “find $f(a)$.”
3	Chain rule	$d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$.	Forgetting to multiply by the inner derivative.	A function raised to a power or nested inside another.
4	Derivative as instantaneous rate	Velocity is the derivative of position; the object is at rest where $v(t) = 0$.	Using position or average velocity instead of $v(t)$.	A position function $s(t)$ and “at $t = \dots$ ”
5	Optimization and the second derivative test	Set $f'(x) = 0$ to find critical points; $f'' < 0$ gives a local maximum and $f'' > 0$ a local minimum.	Mixing up the sign test, or dropping a constraint.	Maximize or minimize with a fixed amount of material.
6	Definite integrals and the Fundamental Theorem	$\int_a^b f(x) dx = F(b) - F(a)$ where $F' = f$; integrating a rate gives total change.	Evaluating the integrand or its derivative instead of the antiderivative.	A rate function and bounds of integration.
7	Exponential growth differential equation	$dy/dt = ky$ has solution $y = y_0 \cdot e^{kt}$.	Using simple linear growth or yearly compounding.	“Grows at a rate proportional to its size.”
8	Product rule	$(fg)' = f'g + fg'$.	Multiplying the two derivatives.	Two functions of x multiplied together.

Course scope (public description): This course guides candidates to apply theoretical concepts of calculus to real-world situations, demonstrating a developing mathematical mindset. This course focuses on limits, derivatives, integrals, and differential equations; it also prepares students for Discrete Mathematics. Prerequisites may include an entrance exam that assesses pre-calculus skills, or readiness; alternatively, completion of pre-calculus within the past 3 – 5 years.